

S-1. Buffer Strip Plantation/Bio-engineering Methods

 Forest / Shrub Land Conservation	 Medium	 High	 Flood Flood Flood Flood Flood Flood	 Implementation Time of Measures
Measure Type	Cost	Preferences by Community	Hazards	Implementation Time of Measures

Descriptions

Measures	For the preservation of soil, water, and biodiversity in hilly areas, buffer strip plants are essential. There are several ecological advantages to these carefully positioned vegetation strips, which serve as a buffer between natural ecosystems and agricultural land. We may establish strong buffer strips that prevent soil erosion, improve water quality, and conserve biodiversity by emphasizing species diversity, employing effective management techniques, and engaging local communities. These kinds of conservation initiatives are crucial to preserving the wellbeing of our ecosystems and the livelihoods of those who rely on them as our environmental problems grow.								
	Bangladesh's CHT are a unique ecological zone with rich cultural legacy, varied flora and wildlife, and green hills. Unfortunately, unsustainable farming methods and deforestation are the main causes of these regions' serious environmental problems, which include soil erosion, deforestation, and biodiversity loss. The establishment of buffer strip plantings in this area provides a feasible conservation strategy to improve environmental resilience and sustain regional economies.								
Location	R-114	R-99	R-65	K-98	K-76	K-71	B-173	B-77	B-76
	■	■	■	■	■	■	■	■	■
Applicable Sites	• Agricultural Lands Adjacent to Water Bodies • Slopes Prone to Soil Erosion • Riparian Zones (Riverbanks & Stream Edges) • Roadside and Settlement Perimeters								
Slope	Less than 3%		3% to 15%		15% to 30%		30% to 60%		Above 60%
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Design options and performance	• Grass Buffer Strips (Vetiver, Napier Grass) – Controls erosion, slows runoff, and improves soil retention. • Agroforestry Buffer Strips – Integrates fruit trees, timber species, and understory crops for dual conservation and income benefits. • Mixed Vegetation Strips (Trees, Shrubs, and Ground Cover Plants) – Provides habitat diversity, windbreak protection, and soil stabilization. • Riparian Buffers (Native Aquatic and Riparian Species) – Filters agricultural pollutants, reduces streambank erosion, and enhances water infiltration. • Contour Buffer Strips – Planted along contours to slow down water runoff and increase infiltration.								
Feasibility criteria	Technical Feasibility								

	• Site Selection & Soil Compatibility – Ensures buffer strips are established in erosion-prone zones. • Selection of Native & Adaptive Species – Uses local plants suited to soil, water, and climatic conditions. • Water Retention & Runoff Management – Optimizes rainwater infiltration and drainage efficiency. Environmental Feasibility: • Soil Erosion Prevention & Land Rehabilitation – Strengthens slope stability and reduces topsoil loss. • Biodiversity Conservation – Enhances wildlife habitat and plant diversity. • Water Quality Improvement – Reduces sediment, nutrient runoff, and contamination of water bodies. Economic Feasibility • Low-Cost Implementation Using Local Resources – Utilizes native plants and community labor. • Multiple Livelihood Benefits – Provides timber, fodder, medicinal plants, and sustainable agroforestry products. • Long-Term Cost Reduction in Land & Water Management – Prevents land degradation and flood-related damages. Social Feasibility • Community Participation & Ownership – Engages local stakeholders in conservation efforts. • Enhanced Food & Income Security – Supports farmers by integrating buffer strips with productive agroforestry species. • Institutional & Policy Alignment – Connects with local and national conservation initiatives.			
Operation and maintenance	• Regular Weeding & Mulching – Maintains soil moisture and prevents invasive species growth. • Replanting & Gap Filling – Ensures consistent buffer coverage and growth. • Drainage & Water Flow Management – Maintains natural runoff channels to prevent waterlogging. • Community-Led Monitoring & Capacity Building – Encourages local training on buffer strip maintenance.			
Cost	Medium			
Benefits				
	Type	Low	Medium	High
	Canopy Cover Increase			
	Carbon Sequestration			
	Crop Production Increase			
	Crop Failure Risk Reduction			
	Ensuring Food Security			
	Fruit Production Increase			
	Forestry Goods Production Increase			
	Livestock Production Increase			
	Fisheries Production Increase			
	Alternative Livelihood Generation			
	Increase of Income			
	Soil Erosion Control			

	Improved Soil Health			
	Soil Fertility Increase			
	Sustainable Water Supply			
	Water Quality Improvement			
	Water Purification			
	River Bank Erosion Control			
	Surface Runoff Reduction			
	Groundwater Recharge			
	Pest and Disease Management			
	Reduction of Pesticides and Fertilizer Use			
	Ecosystem and Biodiversity Increase			
	Better Communications			
	Better Education			
	Better WASH facilities			
Implementation Challenges				
	Type	Low	Medium	High
	Land unavailable/high land area required			
	Supply chain disconnection			
	Land right related problems			
	High human investment capital need			
	Insufficient infrastructures			
	Less demand of agroforestry product			
	Environmentally unsustainable			
	Socially unacceptable			
	Less beneficial			
Environmental performance	- Improved Soil Stability & Reduced Erosion – Reduces runoff intensity and prevents landslides. - Enhanced Water Quality & Hydrological Balance – Filters pollutants before they reach water bodies. - Biodiversity Restoration & Habitat Connectivity – Provides corridors for wildlife movement. - Increased Carbon Sequestration & Climate Adaptation – Strengthens ecosystem resilience and mitigates climate impacts.			
Remarks	This measure will aid landscape restoration in CHT of Bangladesh along with creating alternative livelihood opportunities.			